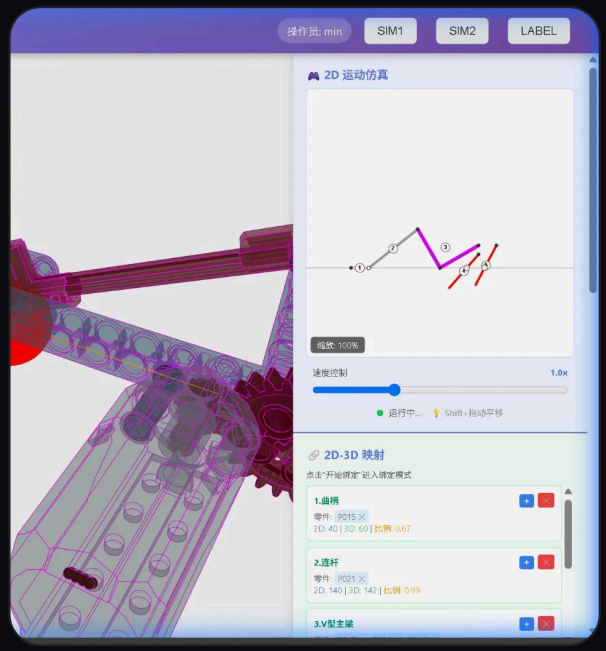
What if any AI agent could understand hardware?

AI agents can trade, code, and write.
But none of them can analyze a mechanical design. Until now.



MakerKit Physics Simulator — Real output

The Knowledge Gap

AI Agents Today

Can trade tokens, search the web, write code, generate images...

Great at digital tasks

≠

Physical Hardware

Gears mesh? Motor strong enough? Chain transfers force?

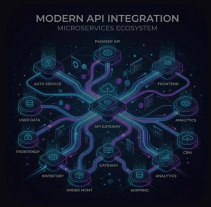
No agent can answer this

Evaluating mechanical feasibility requires **expensive domain expertise** that is slow, inaccessible, and **not available as an API**.

Design Intelligence as a Service

One API call. Any agent becomes a hardware expert.

```
// Agent A wants to evaluate a design POST /analyze // x402 micropayment attached {
"prompt": "A robotic arm that picks up cups", "components": ["motor_28BYJ",
"gear_worm", "arm_segment"] } // CoM Agent returns: { "mechanicalPath": ["Motor → Worm
Gear → Arm Joint → Gripper"], "complexityScore": 576, // Structure × Electronics ×
Logic "feasibility": "PASS", // Physics simulation result "metricsHash": "0x1f2e..."
// On-chain verifiable }
```



How It Works

🤖 **Agent A**
Needs design analysis

→

💳 **x402**
$0.01 micropayment

→

🧠 **CoM Agent**
Analyze in TEE

→

📊 **Result**
Score + Hash

→

⛓ **On-chain**
Hash committed

Agent discovers agent → pays agent → uses result.

No human needed. Pure machine economy.

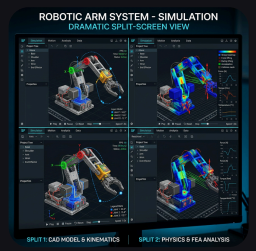
The Recipe Stays Sealed

What Only CoM Agent Knows

- How to trace a mechanical transmission path
- How to score component interactions
- How to predict feasibility from structure
- How to evaluate n^3 complexity

ChatGPT, Claude, Gemini — none of them can do this.

→



What You Get

Score: **576**

Path: Motor → Gear → Arm

Hash: 0x1f2e...

Result, never the recipe.

Lit Protocol TEE ensures the algorithm runs in a sealed environment. IPFS-deployed. Tamper-proof.

Real Hardware Results

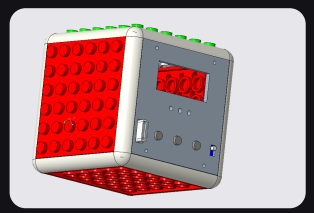
CoM Agent analyzes real MakerKit assemblies — not toy problems.

Modular Parts

Gears, motors, axles, sensors — real components with real physics

Live Simulation

Physics engine validates transmission chains in real-time



Shipping Product

MakerKit Cube — the hardware these designs run on

Built for CROPS

C

Censorship Resistance

Listed on decentralized agent markets (OpenServ, Olas). No single platform controls access.

O

Open Source

API spec is fully open. Anyone can build a competing analysis engine. The interface is the standard.

P

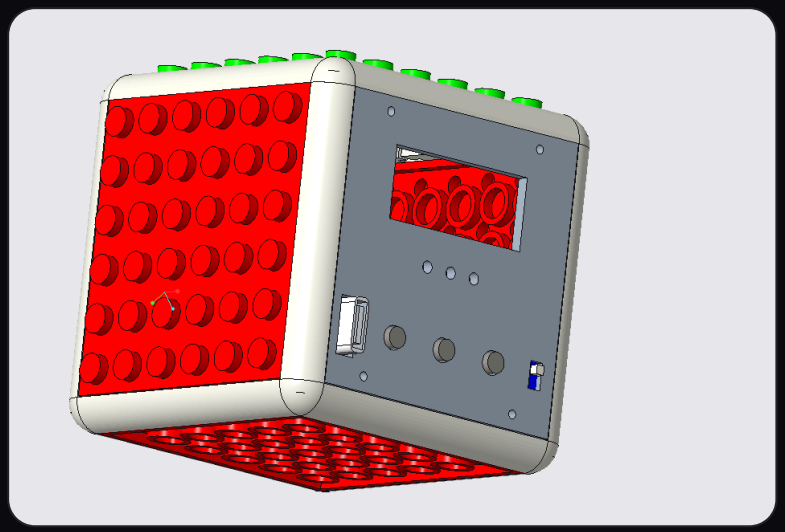
Privacy

Analysis runs in Lit TEE or locally. User prompts never leave their environment. Venice for LLM inference.

S

Security

metricsHash lets consumers verify output correctness without trusting the agent. Graceful degradation if offline.



MakerKit Cube — Real hardware product

The world's first hardware intelligence agent.

Foundation models know what's public.

CoM Agent knows what isn't.

$0.01	<1s	∞
per analysis	response	availability

Wu Xi · Agent #34679 · Harness: Gemini · github.com/wu1wu/com-agent